MAAI: Voice-first Maternal Mental Health Support

One-pager | Bolna x YC Symphony Buildathon | Submission deadline: 31 Oct 2026 | Prepared 18 Oct 2026

1. The idea

MAAI is a multilingual voice agent that calls new and expecting mothers (with their consent), asks the standard EPDS screening questions in Hindi / Hinglish, and passes high-risk cases to a human clinician. It is **screening, support and referral. It is not diagnosis or treatment.** Pitch: *"Many new mothers struggle silently. MAAI checks in with a simple phone call and connects them to human care when needed. No app. Just a conversation."*

2. Who is already doing something similar

Player	What they do (1-2 lines)	Link
Novoflow (US)	AI phone call about 2 weeks after delivery asks EPDS-style questions and flags a nurse / social worker if risk is high. **Closest to MAAI.** Vendor's own page; no Indian languages mentioned.	ai.novoflow.io
Hippocratic AI (US)	Reported to have a postpartum mental health assessment and depression screening agent (made with a maternal mental health group). Secondary source, verify before quoting.	secondary source
Videra Health	Voice and language model predicts EPDS score from about 1 minute of speech (AUC 0.886, 275 women, 2026 peer-reviewed study). Screening only.	viderahealth.com
VillageFor + Canary Speech	Maternal app with EPDS screening, mood check-ins and voice journal; Canary voice analysis added on 8 Oct 2026. App-based, not outbound calls.	prnewswire.com
Woebot (WB001)	CBT chatbot for postpartum depression with FDA Breakthrough designation (2021). Consumer app shut in June 2025; now enterprise partners only.	mobihealthnews.com
Penn Medicine + Memora	Text-based postpartum chat agent that includes PPD screening; 290 patients used it in the first 6 months.	ai.jmir.org
China AI follow-up calls	Research on AI voice assistants making postpartum follow-up phone calls in hospitals.	ncbi.nlm.nih.gov
ARMMAN: Kilkari / MaMitra (India)	Free voice calls to millions of mothers across India, plus a Hindi/Marathi AI chatbot. Gives health information; no PPD screening found. Best seen as a partner / distribution channel.	armman.org | Google blog
Project Saathi (India)	Vernacular voice AI that helps women interpret symptoms and find care, including maternal health. Not PPD specific.	indiaai.gov.in
"Sakhi" (MIT Solve entry)	WhatsApp maternal health support for women served by ASHAs. **Avoid this name.**	solve.mit.edu

Gap (from our limited search, not proof): we did not find an Indian-language, outbound-call PPD screening agent with human escalation. Our edge: Hindi / Hinglish / Punjabi on a normal phone call, ASHA or hospital onboarding, local human referral. Do not claim "first ever".

3. What we will build (MAAI, as per the attached plan)

1. Proactive voice check-ins. Consent-based calls, daily or weekly, in Hindi, English, Hinglish.	**2. EPDS screening by voice.** 10 questions asked in conversation; scoring done by fixed code, not by the LLM.	**3. Wellbeing trends.** Weekly trend from consented answers. No voice-tone diagnosis.
4. Human escalation. Concerning answers go to a trained clinician. Self-harm, infant-harm or psychosis warning signs trigger emergency guidance.	**5. Clinician dashboard.** Screening history, trends, follow-up status. Family is told only with the mother's permission.	**Model: Hybrid.** Mother companion calls + hospital / ASHA dashboard. Hospital or ASHA registers the mother (builds trust).

4. Pipeline / structure

1. Register + consent ASHA / hospital adds the mother and her consent in a web form. Saved in Supabase.	**2. Trigger the call** Scheduler calls the Bolna API to start an outbound call from a Plivo / Exotel number.	**3. Live conversation** Sarvam / Deepgram (speech to text) > LLM with safety rules > Sarvam / Cartesia (text to speech).
4. Capture answers Agent sends each EPDS answer to our FastAPI webhook. Code calculates the score.	**5. Save risk level** Low / medium / high stored in Supabase with call history and transcript.	**6. Act** High risk: alert clinician + emergency guidance. Everything shows in the Next.js dashboard.

Stack: Bolna (orchestration) | Sarvam / Deepgram / Cartesia (speech) | Python FastAPI | Supabase | Next.js + Tailwind | Vercel.

5. How to start: Bolna or ElevenLabs

Option A: Bolna (recommended) Buildathon host. Built for phone calls in 10+ Indian languages, with API triggers and human escalation. Sponsors Sarvam, Plivo, Exotel give the full India stack. **Steps:** (1) Create Bolna account and join their Slack; ask about buildathon credits. (2) Write the agent prompt: consent, EPDS one question at a time, safety rules, hand-off. (3) Call your own phone in Hindi / Punjabi and fix wording. (4) Add a webhook to FastAPI for answers and call status. (5) Build the dashboard on top.	**Option B: ElevenLabs Agents** Strong voice quality, tool calls through MCP / API integrations, and an endpoint to register Twilio calls. Not a buildathon sponsor, and Indian phone setup is more work. **Use it only if** a Bolna voice sounds weak in your language. First test a few Hindi / Punjabi voices in ElevenLabs, and check Bolna docs to see if it can be plugged in as the voice provider. **Rule:** build the full flow on Bolna first; optimize the voice later.

6. Timeline to 31 Oct

18-20 Oct voice agent + consent + test calls | **21-23** EPDS scoring + safety tests | **24-26** dashboard | **27-29** end-to-end demo calls with synthetic users | **30-31** demo video, live deploy, pitch deck, submit.

Safety: screening prototype only. No real patients until consent, clinical review, privacy safeguards and a reliable human emergency escalation exist. Use synthetic or consenting volunteer data in the demo.